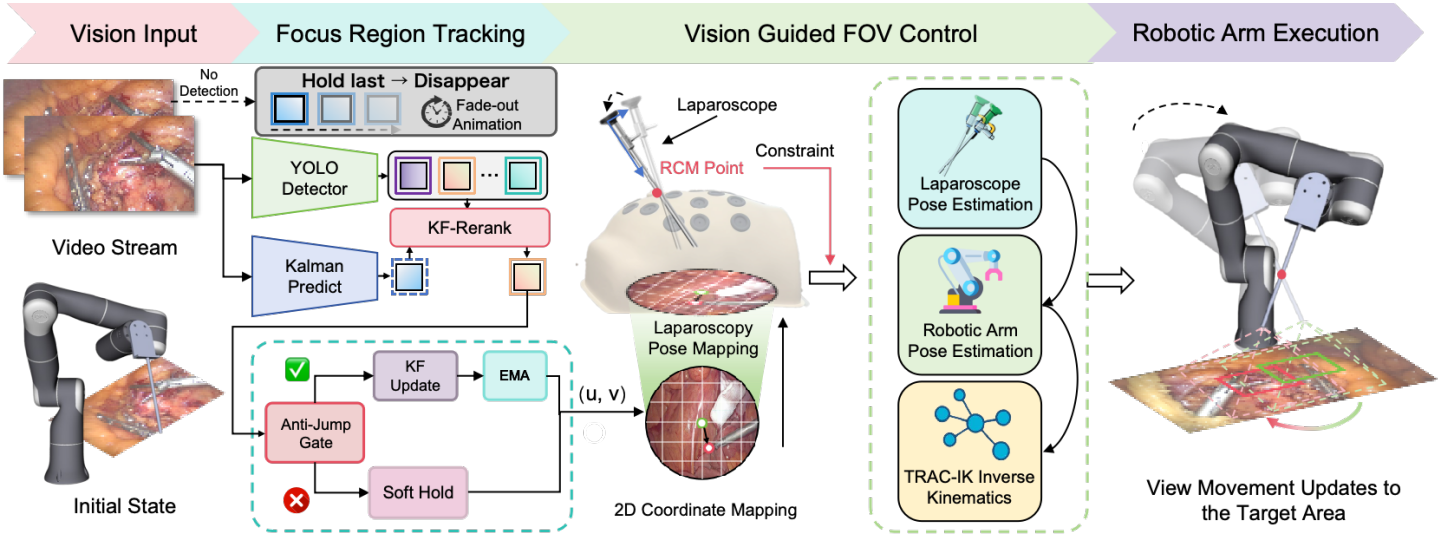
Vision Input
Focus Region Tracking
Vision Guided FOV Control
Robotic Arm Execution
No Detection
Hold last → Disappear
Fade-out Animation
YOLO Detector
KF-Rerank
Kalman Predict
Video Stream
Initial State
KF Update
EMA
Anti-Jump Gate
Soft Hold
(u, v)
Laparoscope
RCM Point
Constraint
Laparoscopy Pose Mapping
2D Coordinate Mapping
Laparoscope Pose Estimation
Robotic Arm Pose Estimation
TRAC-IK Inverse Kinematics
View Movement Updates to the Target Area